# Uchaaran

## A little practice. A deeper connection.

Uchaaran Helper · Kids Mode for Sundar Gutka

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### The problem we are solving

Children need help pronouncing Gurbani and a reason to return to daily practice. Conversations with Khalis focused our design on engagement that sustains practice, and understanding that goes beyond repeating words.

## Two modes. One path toward mastery.

### 1 Practice at your pace

Listen and repeat with Gurmukhi, Roman-letter pronunciation and English meaning. Use recall mode to practice from memory.

### 2 Take the pauri challenge

Recite a line and tap Done. Azure returns word-level feedback. Revisit difficult words across 38 pauris; stars and Nanak coins unlock character rewards.

### What changed since midterm

Organizers asked for less pressure and simpler feedback. We added line-by-line scoring, short word prompts, and an ML maintenance guide for Khalis.




### Rooted in the Panth

Built for children, initially grades 6 to 8. Gurmukhi and the ustaad's recitation stay central. Familiar landmarks form a memory palace for recall; Sikh-inspired characters and seva stories connect learning with community.

### Private processing, by design

Recordings are processed privately in Azure. Temporary server recordings are deleted after scoring, with no use for public-model training.

Khalis can host the model in its own shared AWS or Azure environment. On-device packaging is another option for fully local processing that keeps recordings on device.

### Tools and implementation

Expo · React Native · TypeScript · Azure-hosted IndicConformer · BaniDB · reference audio. Codex and Paper supported development and design.

**Ready for handover.** MIT source, setup and ML maintenance docs included. Khalis supplies hosting and credentials; costs depend on usage. An ongoing technical owner remains to be named.

Watch demo ↗ | Try the app ↗ | View code ↗ | Design process ↗